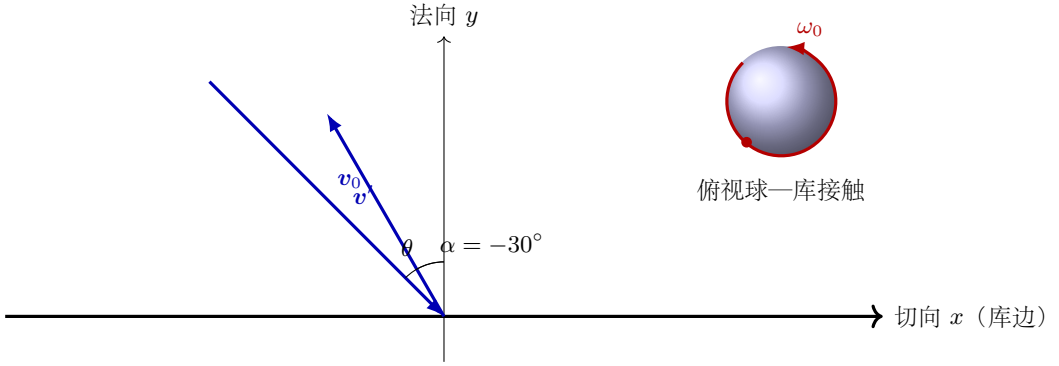
法向 $y$
$\omega_0$
俯视球—库接触
$\boldsymbol{v}_0$
$\boldsymbol{v}$
$\theta$
$\alpha = -30^\circ$
切向 $x$（库边）